Lab 1

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Experiment 1

Measure the average and peak-to-peak inductor current, and the average output voltage. Compare with your pre-lab values and account for any deviation.

I_L (Average inductor Current)	**Δi_L (Peak to Peak Inductor current**	**V (Average output Voltage)**
3.91 A	2.25 A	30.046 V

Quantity	*Prelab values*	*Measured*	*Deviation*
I_L	4 A	3.91 A	*-2.3%*
Δi_L	1.28 A	2.25 A	*75.8%*
V	32	30.046 V	*-6.1%*

Is the converter in CCM or DCM? Justify from the waveform.

The converter is operating in CCM. The buck converter outputs a full triangle waveform.

From our measured values:

$I_L > \Delta i_L/2$

3.91 A > 2.25 A / 2

3.91 A > 1.125 A

Compute the power transfer efficiency from the BK front-panel readings and the ITECH's voltage and current readings. Identify the loss mechanisms.

Efficiency

P_{out} = 30.45 V × 3.807 A = 115.9 W

P_{in} = 40.7 V × 3.1 A = 126.2 W

η = 115.9W / 126.2W = 91.86%

Loss mechanisms: IGBT conduction and switching losses, diode conduction and reverse recovery losses, inductor copper and core losses, capacitor ESR loss, and cable resistance.

Experiment 2

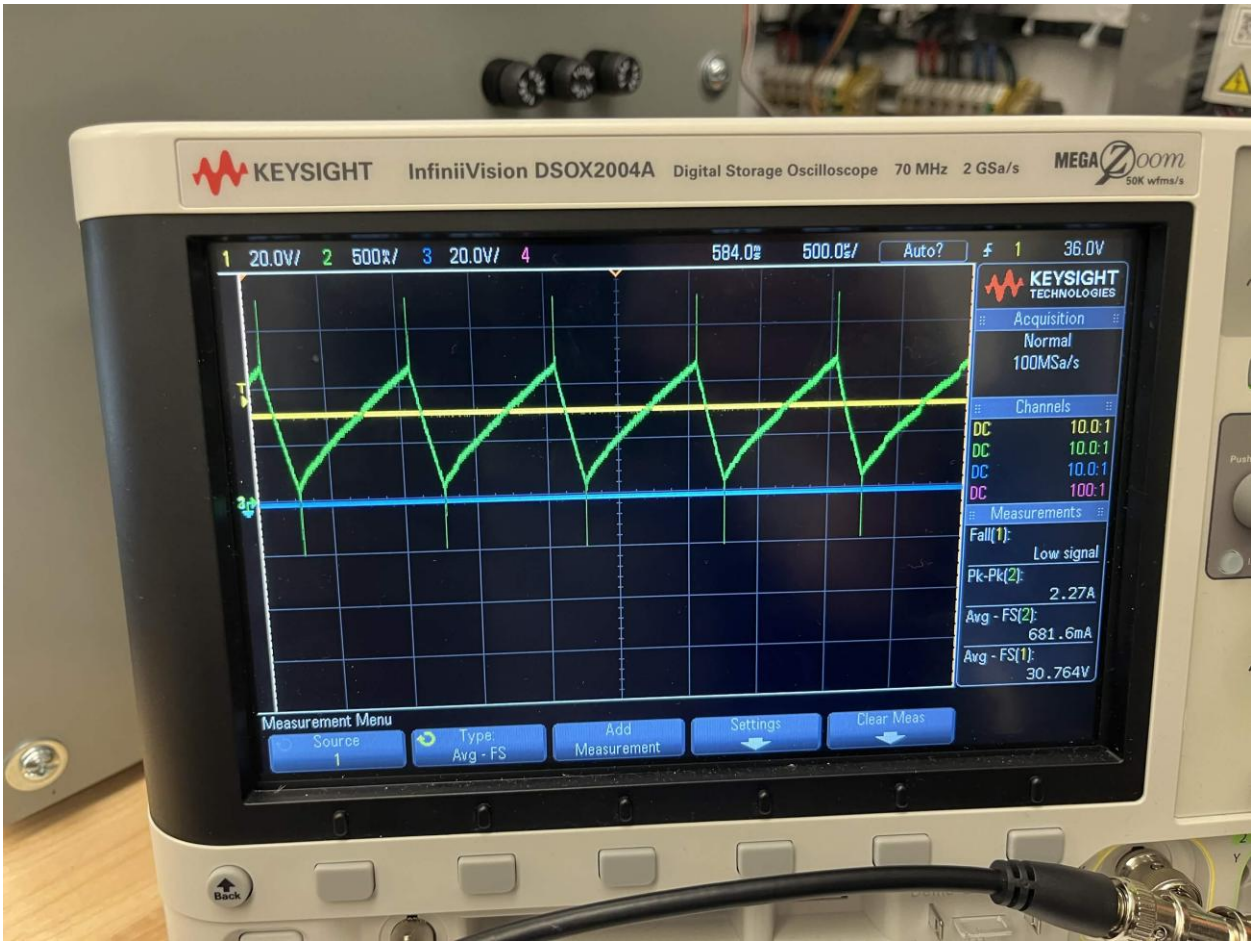


Figure 1

Report the load current and equivalent resistance at the boundary and compare with your pre-lab prediction. What does the boundary current tell you directly about the ripple?

I_L (measured)	Δi_L (measured)	V (measured)	R	I_L (prelab)	Vo (prelab)
3.91 A	2.25 A	30.046 V	8 Ω	4.0 A	32 V
2.54 A	2.3 A	30.46 V	12.5 Ω	2.56 A	32 V
1.28 A	2.3 A	30.6 V	25 Ω	1.28 A	32 V
0.635 A	2.3 A	30.81 V	50 Ω (boundary)	0.64 A	32V
0.345 A	1.5 A	33.5 V	100 Ω	0.32 A	35.2 V

The ripple stays roughly the same until the converter enters DCM. As load resistance increases, the average inductor current decreases until it reaches the CCM/DCM boundary, which in our case is 50 ohms. At this boundary, half the peak-to-peak ripple ($\Delta iL/2$) equals the average inductor current, meaning the inductor current's minimum just touches zero. Any load resistance greater than 50 ohms pushes the average current below this point. Since the current can't go negative (the diode blocks reverse conduction), it instead stays at zero for part of the switching period, which is what defines DCM operation.

Experiment 3

Measure the average output voltage. It will not equal DV_g – explain why

V	R
15.27 V	100 Ω
11.26 V	50 Ω
8.02 V	25 Ω

It doesn't equal DV_g because the conversion ratio is different for a buck converter operating in DCM. There is an interval where both switches are off. A low duty ratio and lighter load can cause the buck converter to operate in DCM.

Annotate the intervals of the inductor current waveform: which device conducts in each, and why does the current stay at zero for part of the period?

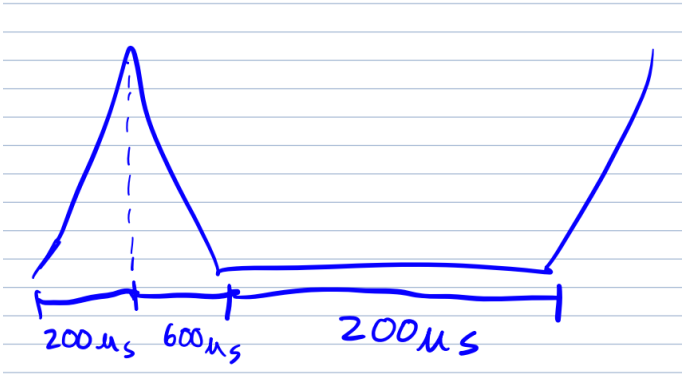


Figure 2

From 0µs ≤ t ≤ 200µs, I_L is rising because Vg is charging the inductor. TOP1 is on and diode is off.

From 200µs ≤ t ≤ 800µs, I_L falls because V_L releases its energy into the capacitor and load. TOP1 is off and diode is on.

From 800µs ≤ t ≤ 1000µs, there is nothing driving I_L because both switches are off. Both TOP1 and diode are off.

The current stays at zero because the diode only allows current to flow in one direction, otherwise there would be negative current.

Experiment 4

Compare the waveforms and average output voltage with Experiment 3. What changed, and why does gating BOT1 eliminate the behaviour you saw there?

V	R
15.27 V	100 Ω
11.26 V	50 Ω
8.02 V	25 Ω

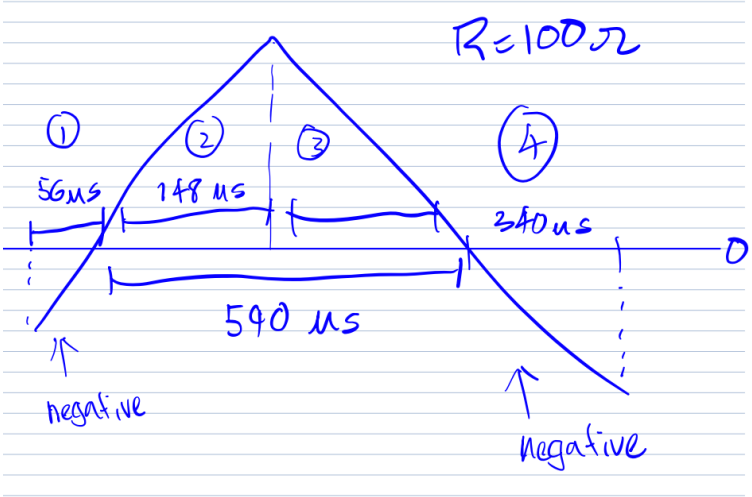


Figure 3

The oscilloscope shows negative current which means it is flowing in the other direction. This is because in experiment 3 we use a diode which only allows current in one direction. In experiment 4, gating BOT1 will make it act as an IGBT, when turned on acts as a wire allowing current to flow in the other direction.

Identify the intervals of negative inductor current. For each gating interval, identify the conducting device for positive and negative current. An IGBT conducts in one direction only, regardless of its gate command.

Inductor Current Waveform Intervals at R = 25

Interval	Duration	Slope / Current	Gate Interval	Conducting Device
1	0 < t < 20us	+ slope, - i_L	TOP1 gated	TOP1 diode
2	20us < t < 200us	+ slope, + i_L	TOP1 gated	TOP1 IGBT
3	200us < t < 860 us	- slope, + i_L	BOT1 gated	BOT1 diode
4	860us < t < 1ms	- slope, - i_L	BOT1 gated	BOT1 IGBT

Experiment 5

Note: Due to a faulty interface board, our Vref yielded different values shown on the oscilloscope experiment 5, 6 and 7. We were limited on time so there are parts of Part B left unfinished due to insufficient lab data.

Capture the steady-state waveforms. Measure the average output voltage, the average inductor current and the peak-to-peak ripple; compare with your pre-lab values.

I_L (Average inductor Current)	**Δi_L (Peak to Peak Inductor current**	**V (Average output Voltage)**	**V_{ref}**
1.55 A	1.55 A	18.35 V	40 V

Quantity	*Prelab values*	*Measured (For V_{ref} = 20 V*	*Deviation*
I_L	6.4 A	6.6 A	*3.1%*
Δi_L	0.4 A	0.8 A	*100%*
V	40	38.35 V	*-4.1%*

Determine the BOT1 duty ratio from the inductor current waveform: the current rises while BOT1 conducts and falls while TOP1's diode conducts, so D is the rising interval divided by the switching period. Check it against the slopes, v_g/L and $(v_g - v)/L$. Compare with the ideal prediction and explain the difference using the circuit's non-ideal behaviour and your measurements.

V_g = 20V
V = 40V
L = 5mH

$T_s = 1/f_s = 1/5000 = 200\ \mu s$
R = 12.5 Ω

Ideal duty ratio

$D_{ideal} = 1 - V_g/V = 1 - 20/40 = 0.50$ (pre-lab)
At the measured output voltage: $D_{ideal} = 1 - 20/38.35 = 0.478$

Measured duty ratio

I = V/R = 38.35/12.5 = 3.07 A
In CCM the diode carries the load current: $I = (1 - D) \cdot I_L$
$1 - D = 3.07/6.6 = 0.465$
D = 0.535

Rising interval = $D \cdot T_s = 0.535 \times 200\ \mu s \approx 107\ \mu s$
Falling interval = $(1 - D) \cdot T_s \approx 93\ \mu s$

Slope check

On-interval (BOT1 on): ideal slope $V_g/L = 4000$ A/s
Off-interval (TOP1 diode on): ideal slope $(V_g - V)/L = (20 - 38.35)/0.005 = -3670$ A/s

With ideal slopes, volt-second balance gives $20 \cdot D = 18.35 \cdot (1 - D)$, so D = 0.478. The measured D = 0.535 is higher. The difference is the voltage lost in the circuit:
$(1 - D) \cdot V = V_g - I_L \cdot R - V_{on}$
$0.465 \times 38.35 = 17.83$ V
Voltage lost = 20 − 17.83 = 2.17 V

So the real inductor voltages are about:
On: $v_L \approx 20 - 2.17 = 17.8$ V → slope ≈ 3560 A/s (shallower than V_g/L)
Off: $v_L \approx 17.8 - 38.35 = -20.5$ V → slope ≈ −4110 A/s (steeper than $(V_g - V)/L$)
Check with volt-second balance: $17.8 \cdot D = 20.5 \cdot (1 - D)$, which gives D = 0.535 ✓

Quantity	Ideal	Measured	Deviation
D	*0.500 (0.478 at 38.35 V)*	*0.535*	*+7.0% (+11.9%)*
Rising slope	*4000 A/s*	*≈ 3560 A/s*	*−11%*
Falling slope	*−3670 A/s*	*≈ −4110 A/s*	*+12%*

Estimate the efficiency: input power from the ITECH voltage and current readings, output power as V_{rms}^2/R (or state the small-ripple approximation if you use the average voltage.) As in Part A, this estimate excludes the separately supplied auxiliary power. Relate the efficiency to the duty-ratio difference above.

(For V_{ref} = 20 V)
$P_{out} = (38.35\ V)^2 / 12.5\ \Omega = 117.7\ W$
$P_{in} = 20\ V \times 6.6\ A = 132\ W$
$\eta = 117.7\ W / 132\ W = 89.17\%$

Loss breakdown
$P_{loss} = P_{in} - P_{out} = 132 - 117.7 = 14.3\ W$

Loss mechanism	**Estimate**	**Power**
Inductor/wiring resistance	$I_L^2 \cdot R = 6.6^2 \times 0.15\ \Omega$	≈ 6.5 W
IGBT and diode on-state drops	$V_{on} \cdot I_L = 1.2\ V \times 6.6\ A$	≈ 7.9 W
Switching, diode recovery, C_{dc} ESR	small at 5 kHz	≈ 1 W or less
Total estimated		≈ 14–15 W
Measured (Pin − Pout)	132 − 117.7	14.3 W

Most of the loss is conduction loss, split roughly evenly between the resistance in the inductor current path and the on-state voltage drops of the BOT1 IGBT and the TOP1 diode. Switching losses are small because the switching frequency is only 5 kHz.

Relation to the duty-ratio difference
In CCM: $I_o = (1 - D) \cdot I_L$, so
$\eta = P_{out}/P_{in} = V \cdot I / (V_g \cdot I_L) = (1 - D) \cdot V/V_g$
$\eta = 0.465 \times 38.35 / 20 = 0.892 \approx 89\%$

Experiment 6

Note: We were unable to get full completion of experiment 6 due to limited time

I_L (Average inductor Current)	**Δi_L (Peak to Peak Inductor current**	**V (Average output Voltage)**	**V_{ref}**
1.55 A	1.55 A	18.35 V	40 V
11.6 A	0.8 A	48.43 V	30 V
8.6 A	1 A	43.41 V	25 V
6.6 A	0.8 A	38.35 V	20 V

Report the rise time, overshoot and settling time of the output voltage. Measure rise time from 10% to 90% of the voltage change; report overshoot relative to the final reference and settling time to a ±2% band around it

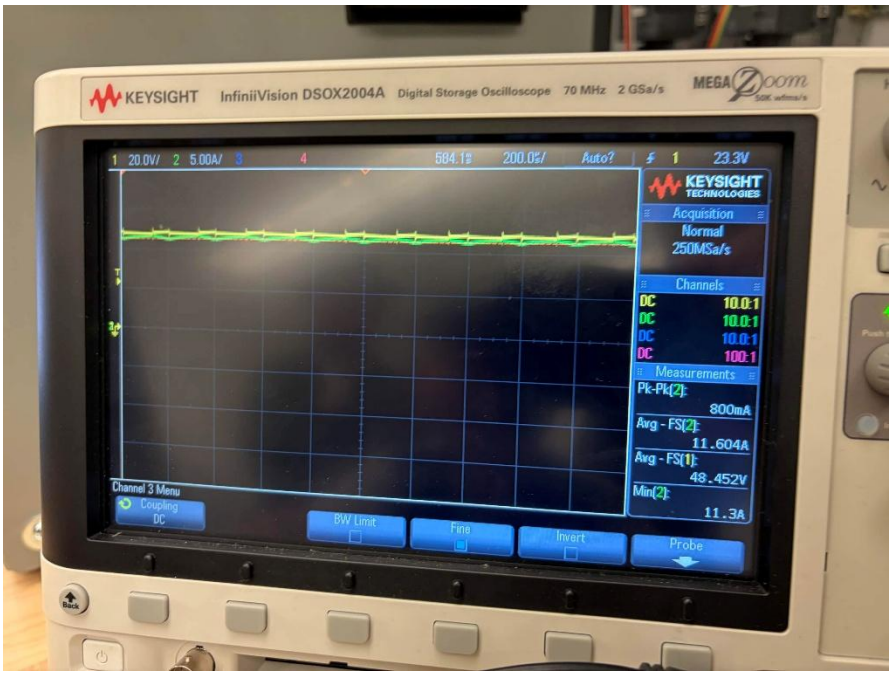


Figure 4 (V_{ref} = 30 V)

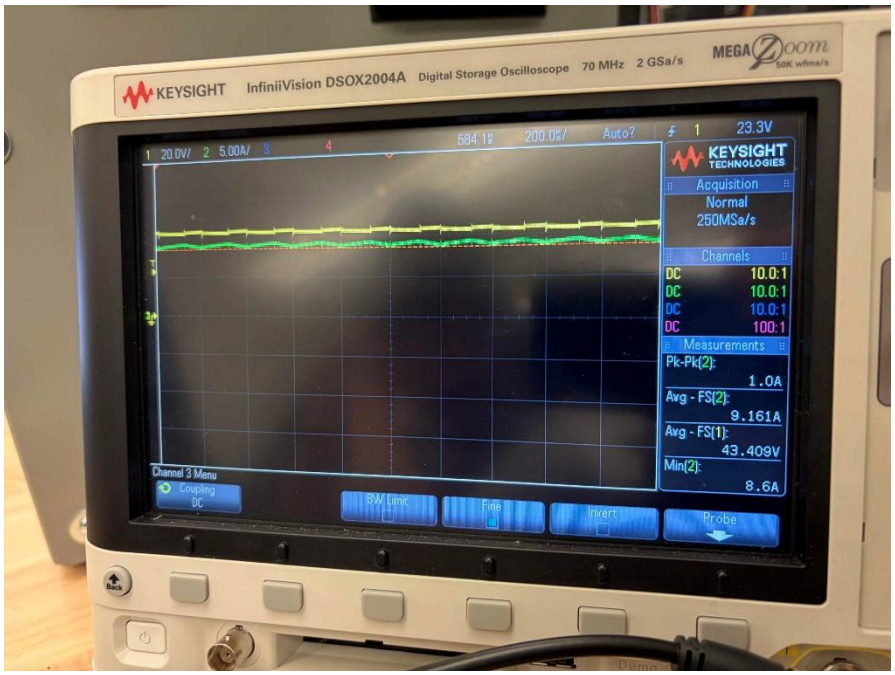


Figure 5 (V_{ref} = 25 V)

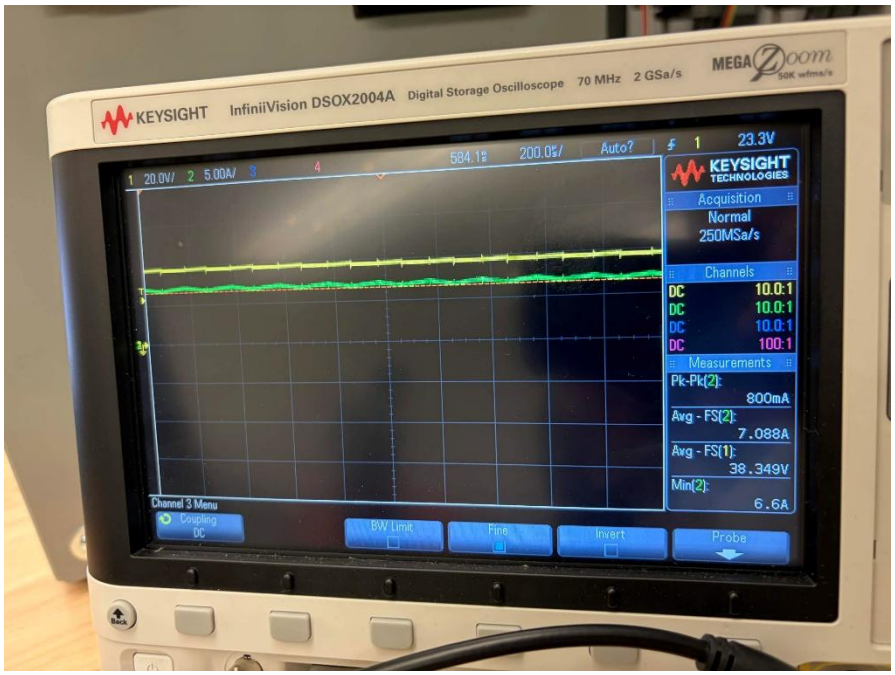


Figure 6 (V_{ref} = 20 V)

From the inductor current, describe what the duty ratio does during the transient. Why must it temporarily differ from its final steady-state value? How close does I get to the 0.7 clamp at 45V, and what would happen to the response if the step were larger?

As inductor current rises, duty ratio also rises. D is different from its steady-state value as volt-second balance and charge balance must hold true to hold the new voltage. We would assume it would get close but not above 0.7 and will engage as the step gets larger.

Experiment 7

Tabulate the average output voltage, the average inductor current, the peak-to-peak ripple and the duty ratio at the three loads. The average current changes with load while the ripple and the duty ratio do not. Explain why, from the inductor volt-second balance and the boost conversion ratio.

Resistance	Δi_L (Peak to Peak Inductor current	I_L (Average inductor Current)	V (Average output Voltage)	Duty Ratio
50 Ω	480mA	1.76A	38.48V	0.52
25 Ω	440mA	3.52A	38.48V	0.51
12.5 Ω	460mA	7.18A	38.39V	0.54

As the load resistance decreases, the average inductor current increased from 1.76A to 7.18A because the converter had to supply a greater power output while keeping the same output voltage. Dropping the resistance by half results in doubling the current. The duty ratio remains approximately the same, as well as the ripple current. From the volt-second balance equation, the boost conversion ratio depends mainly on V_g, V_o and D, independent of the load resistance. The current ripple depends on V_g, L, f_s, and D. Since these components remained mainly constant, the ripple and duty ratio barely changed. The tiny changes are due to real converter losses.

Would the output voltage stay at 40 V across these three loads if the loop were disabled and the duty ratio frozen? Answer the ideal converter and then for the real one, naming the effect that separates them.

For an ideal boost converter, the output voltage would remain at 40V if the duty ratio was frozen at D = 0.5. $V_o = V_g / (1-D)$ is independent of load resistance, meaning output voltage will not change. In a real converter, the output voltage would vary with load because of conduction losses and parasitic voltages, which increases with current. This effect is “load regulation”, where the output voltages changes as the load changes for a fixed duty ratio. The controller’s closed-loop feedback fixes this effect by adjusting the duty ratio. We can see this with our data, as D changes as resistance is changed.